

# Supplementary Material

## Before Scientific Emergence Becomes Visible: Independent Evidence for Structural Precursors in Large-Scale Scholarly Knowledge

Mohammad Amir Khusru Akhtar

### Supplementary scope

This supplement preserves the complete software output associated with the frozen PBSK v4.0.0 run. Every software-generated figure is reproduced below. Every software-generated CSV table is included in full as a machine-readable file under `supplementary_data/results_tables/`; the PDF provides an inventory and a readable preview for each table. Empty tables are retained and explicitly identified rather than silently omitted. The supplement therefore separates human-readable presentation from the exact machine-readable computational record.

Inventory of all software-generated result tables supplied with this submission.

| ID  | Software table                                    | Rows  | Columns |
|-----|---------------------------------------------------|-------|---------|
| S1  | table_ablation_openalex.csv                       | 9     | 7       |
| S2  | table_book_operationalization.csv                 | 40    | 7       |
| S3  | table_creative_universe_generalization.csv        | 33    | 6       |
| S4  | table_cross_source_agreement.csv                  | 8     | 5       |
| S5  | table_figure_generation_qc.csv                    | 14    | 3       |
| S6  | table_hypothesis_tests_openalex.csv               | 69    | 12      |
| S7  | table_incremental_value_openalex.csv              | 4     | 8       |
| S8  | table_landmark_benchmark.csv                      | 12    | 8       |
| S9  | table_matched_controls.csv                        | 24    | 6       |
| S10 | table_oof_predictions_openalex.csv                | 1012  | 7       |
| S11 | table_openalex_bookwide_features.csv              | 108   | 50      |
| S12 | table_power_validation_ablation.csv               | 9     | 7       |
| S13 | table_power_validation_controls.csv               | 240   | 17      |
| S14 | table_power_validation_cross_source_agreement.csv | 8     | 5       |
| S15 | table_power_validation_features_openalex.csv      | 1080  | 50      |
| S16 | table_power_validation_features_pubmed.csv        | 354   | 50      |
| S17 | table_power_validation_hypotheses.csv             | 69    | 12      |
| S18 | table_power_validation_incremental_value.csv      | 4     | 8       |
| S19 | table_power_validation_oof_predictions.csv        | 11880 | 7       |
| S20 | table_power_validation_positive_topics.csv        | 120   | 16      |
| S21 | table_power_validation_predictive_performance.csv | 11    | 14      |

Continued on next page

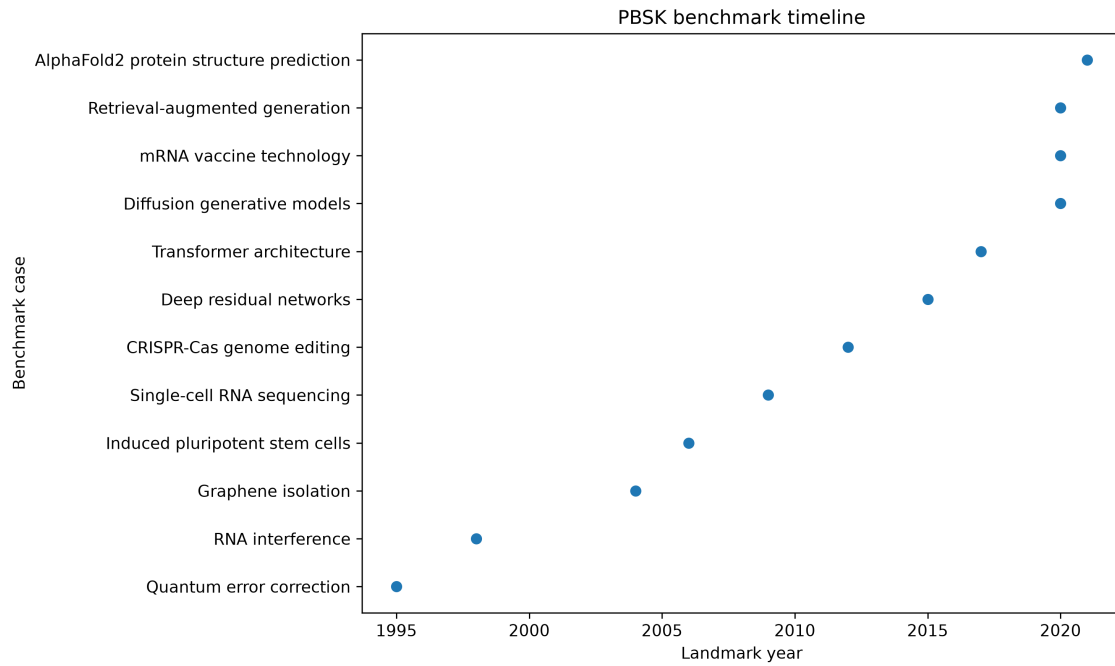
Inventory of software-generated result tables (continued)

| ID  | Software table                                  | Rows | Columns |
|-----|-------------------------------------------------|------|---------|
| S22 | table_power_validation_pubmed_replication.csv   | 0    | 0       |
| S23 | table_power_validation_temporal_performance.csv | 15   | 15      |
| S24 | table_predictive_performance_openalex.csv       | 11   | 14      |
| S25 | table_pubmed_bookwide_features.csv              | 48   | 50      |
| S26 | table_pubmed_replication_hypotheses.csv         | 0    | 0       |
| S27 | table_pubmed_replication_transformation.csv     | 5    | 9       |
| S28 | table_region_features_all_sources.csv           | 1590 | 50      |
| S29 | table_retrieval_qc.csv                          | 1068 | 12      |
| S30 | table_sequence_architecture.csv                 | 6    | 5       |
| S31 | table_temporal_leakage_audit.csv                | 1590 | 6       |
| S32 | table_temporal_predictive_performance.csv       | 15   | 15      |
| S33 | table_transformation_outcomes.csv               | 414  | 15      |
| S34 | table_transformation_tests_openalex.csv         | 5    | 9       |

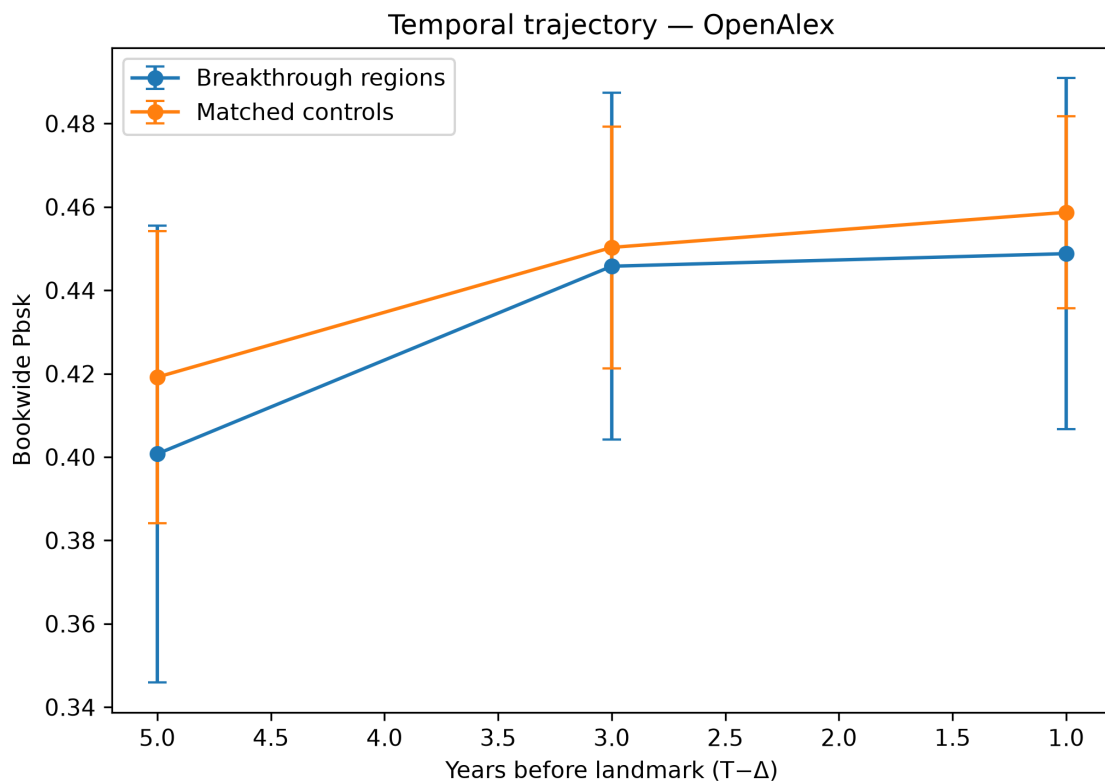
The visual record is reproduced in figures S1 to S13. The analytical record is represented by tables S1 to S11 (historical landmark outputs), tables S12 to S23 (independent power-validation outputs), and tables S24 to S34 (additional predictive, PubMed, quality-control, sequence, leakage, and transformation outputs). Each preview is accompanied by the exact CSV filename, and the complete CSV is retained in the submission package.

## Software-generated figures

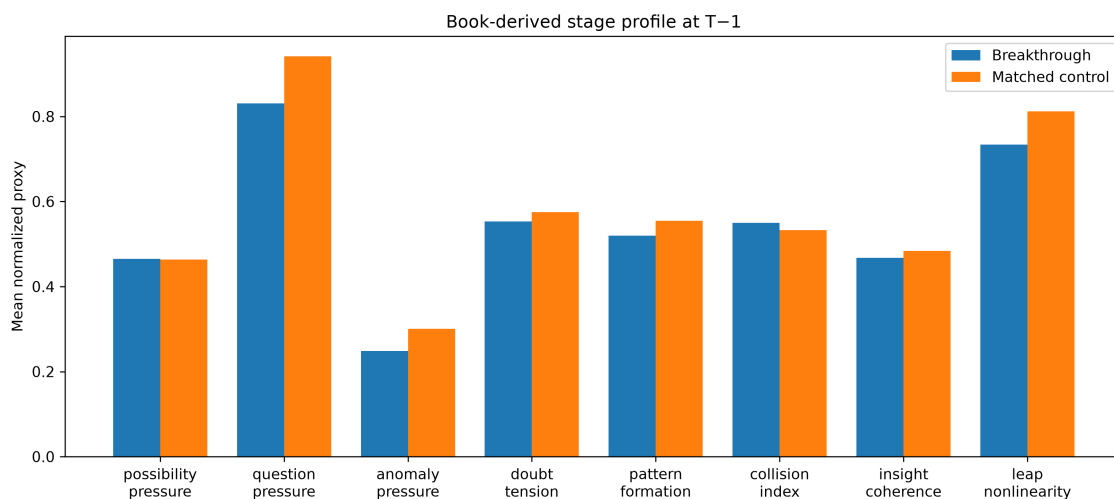
Figures S1–S13 reproduce every original software-generated figure retained in the frozen run. The captions describe the analytical content rather than simply repeating software filenames.



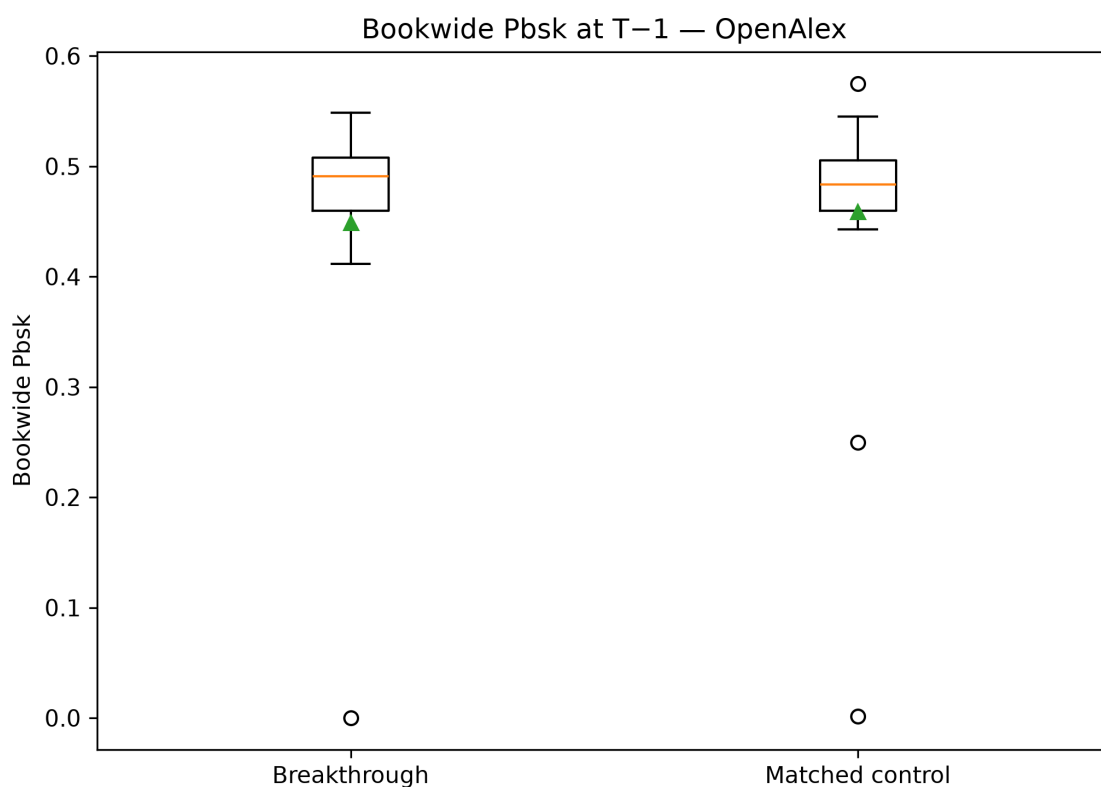
**Figure S1:** Historical benchmark timeline for the 12 landmark cases, showing landmark years used to anchor the exploratory T-5, T-3, and T-1 analyses.



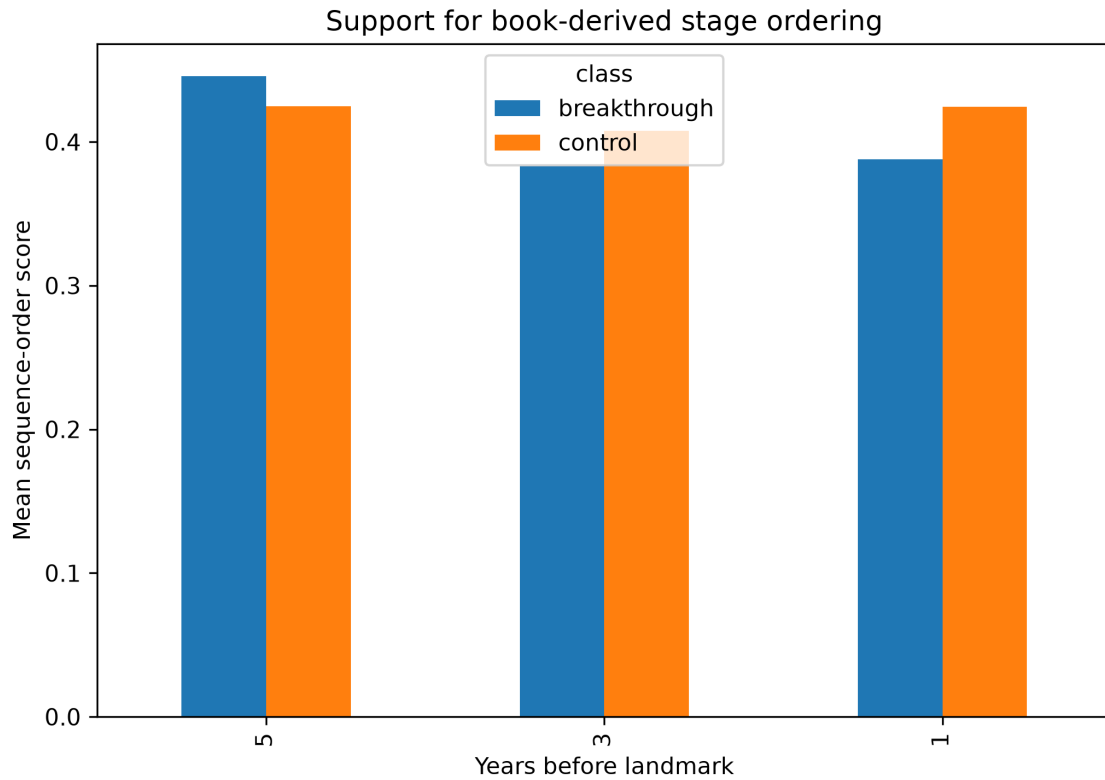
**Figure S2:** Historical OpenAlex trajectory of the book-wide PBSK score for landmark breakthrough regions and matched controls across T-5, T-3, and T-1; error bars show between-region variability as generated by the frozen software run.



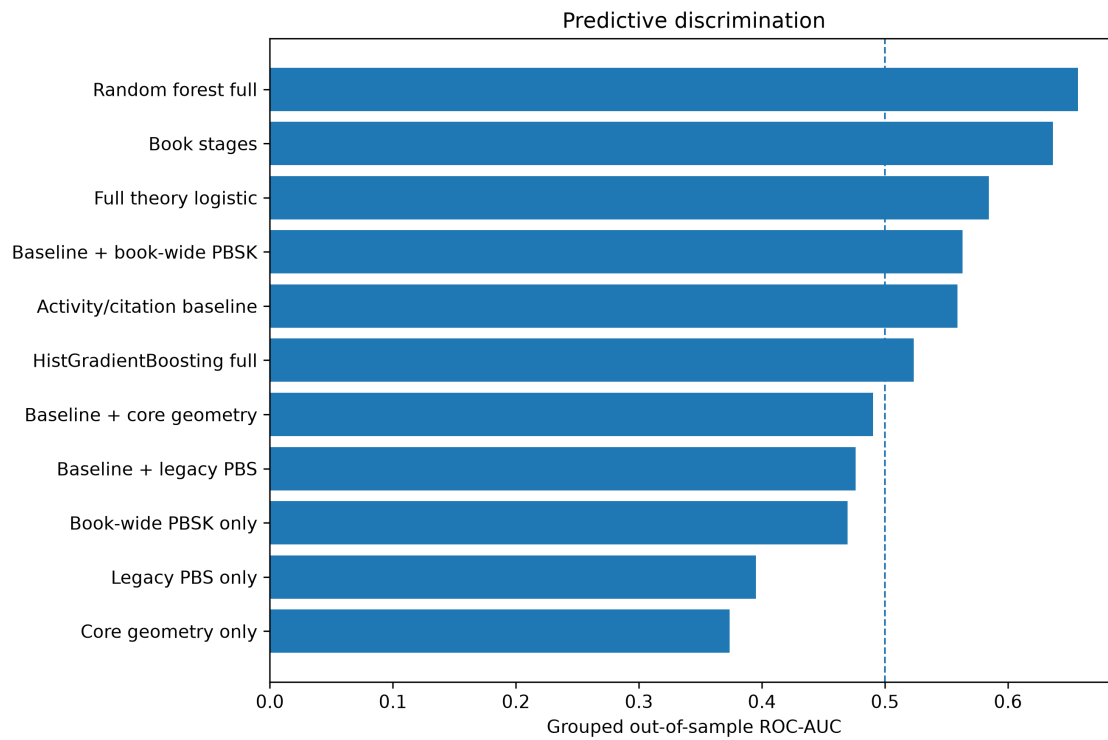
**Figure S3:** Historical OpenAlex profile of selected book-derived proxy measures at T-1 for landmark breakthrough regions and matched controls.



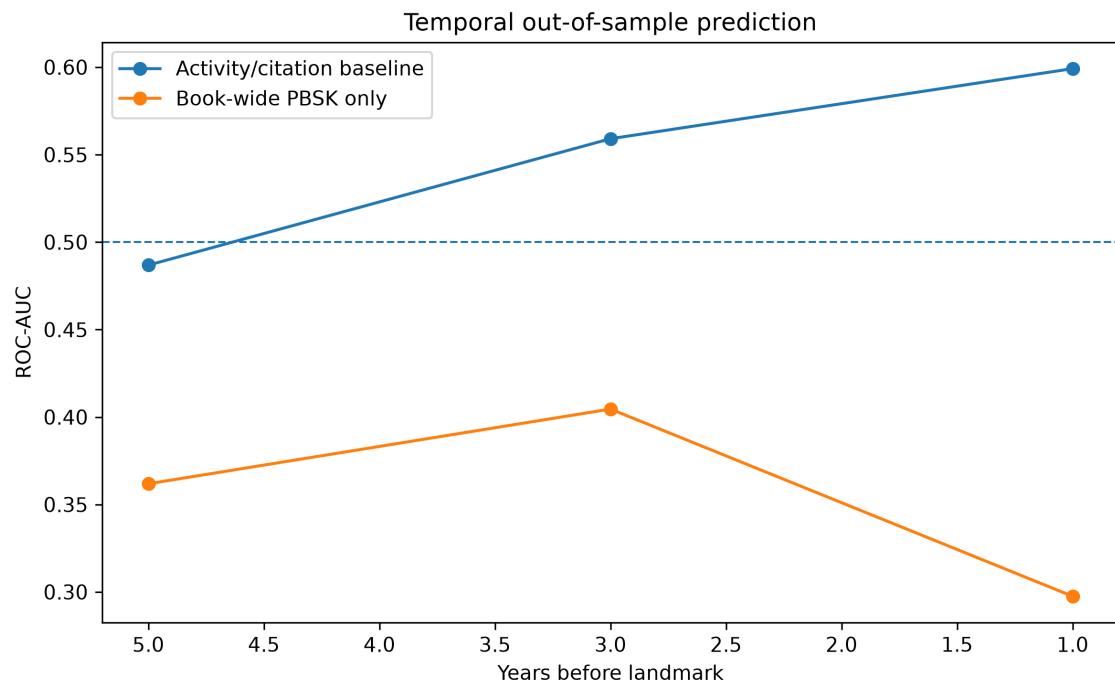
**Figure S4:** Historical OpenAlex distribution of the book-wide PBSK score at T-1 for landmark breakthrough regions and matched controls.



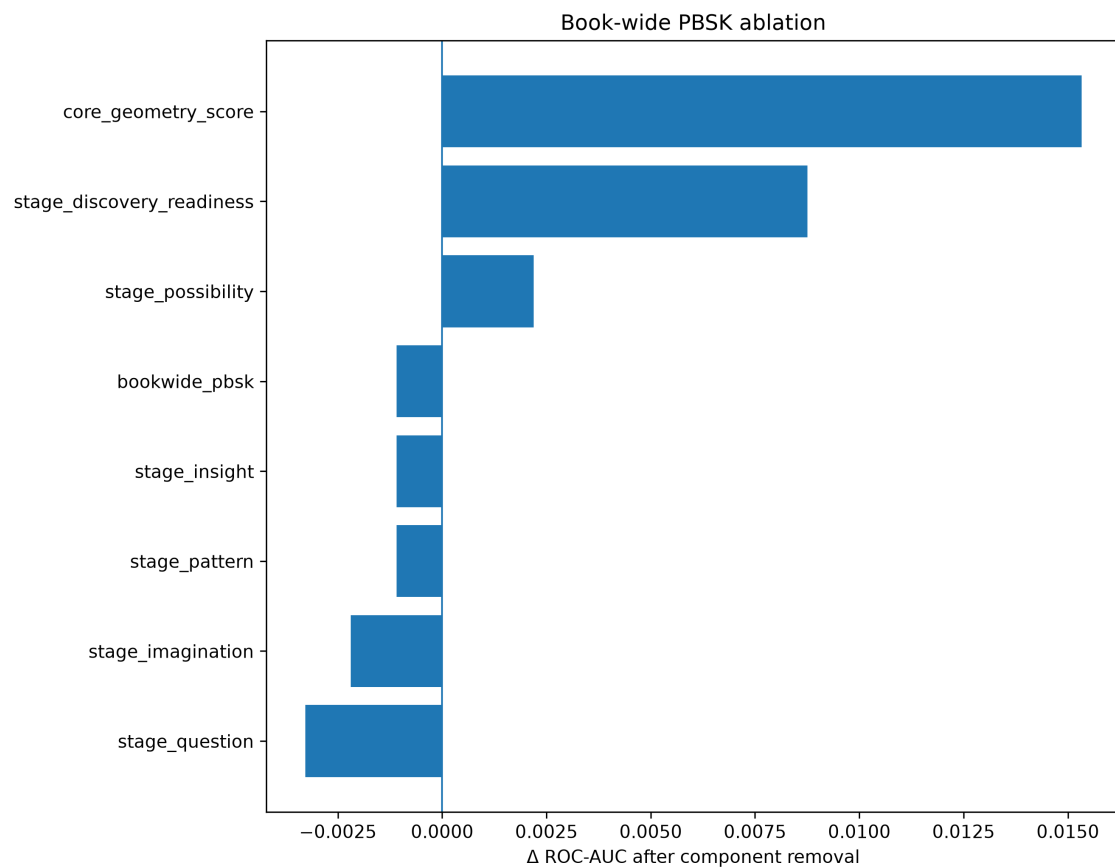
**Figure S5:** Historical landmark support for the book-derived sequence ordering at T-5, T-3, and T-1, shown as mean sequence-order scores for breakthrough regions and controls.



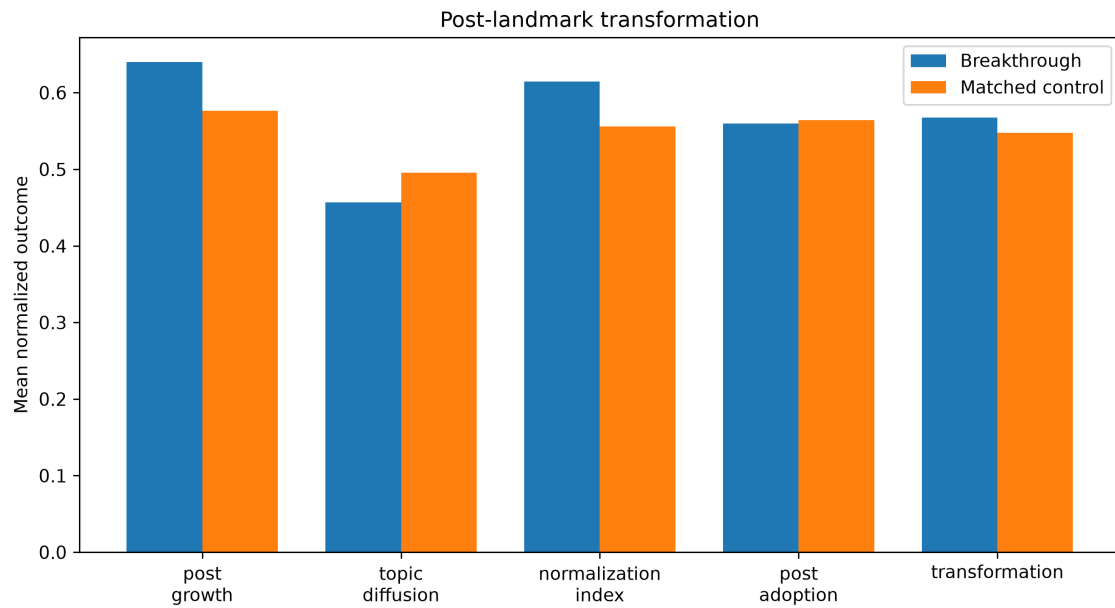
**Figure S6:** Historical landmark grouped out-of-sample ROC-AUC across predictive models. The dashed vertical line marks chance discrimination at ROC-AUC = 0.5.



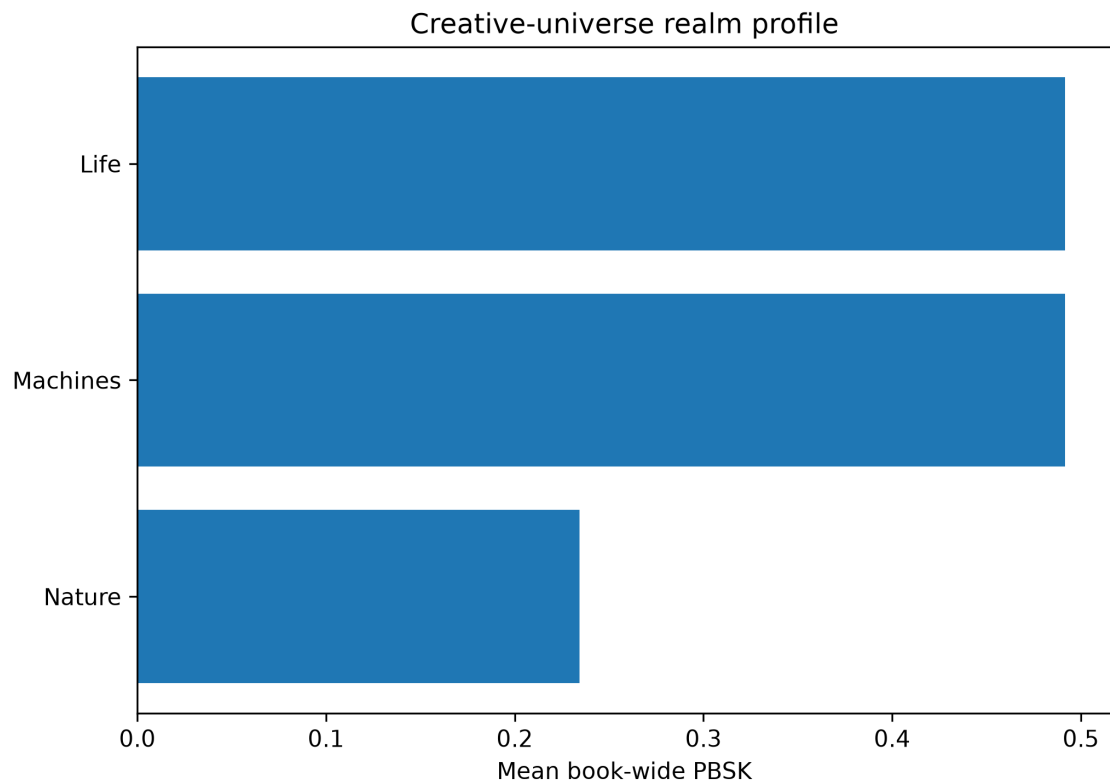
**Figure S7:** Historical landmark temporal out-of-sample ROC-AUC for the conventional activity/citation baseline and the standalone book-wide PBSK score at T-5, T-3, and T-1.



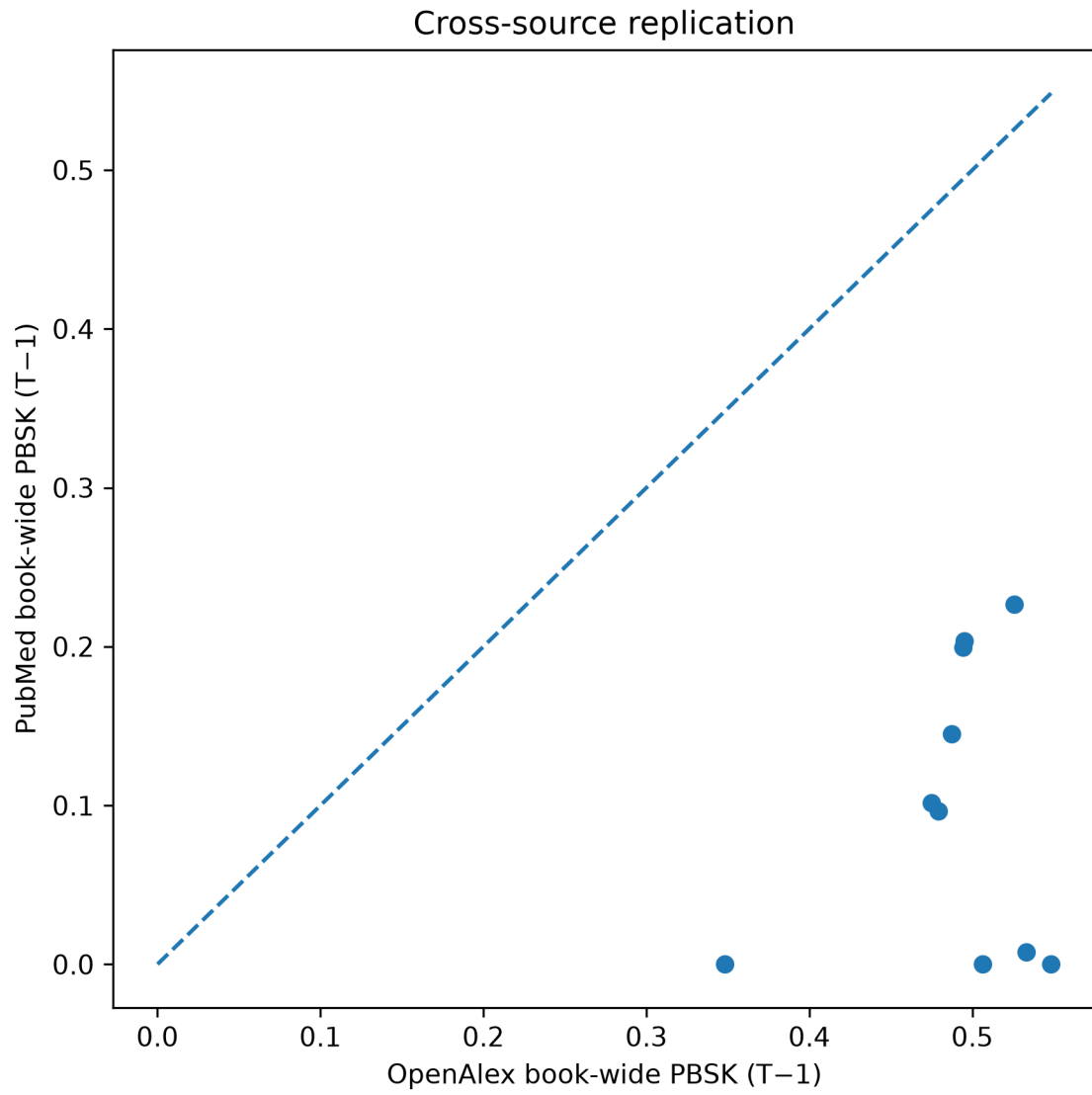
**Figure S8:** Historical landmark ablation analysis. Bars show the change in grouped ROC-AUC after removal of each component from the book-wide model.



**Figure S9:** Historical post-landmark transformation profile, comparing mean normalized growth, topic diffusion, normalization, adoption, and composite transformation outcomes for landmarks and controls.

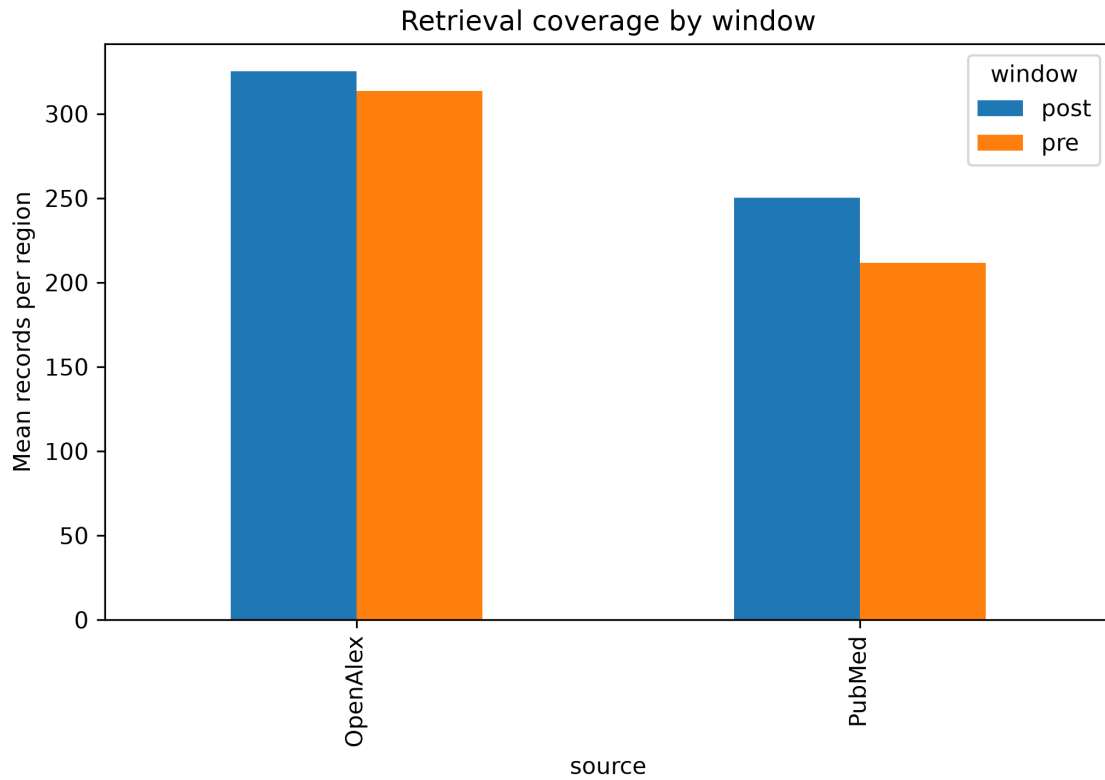


**Figure S10:** Exploratory book-wide PBSK profile across the software-defined creative-universe realms represented in the historical landmark cohort.

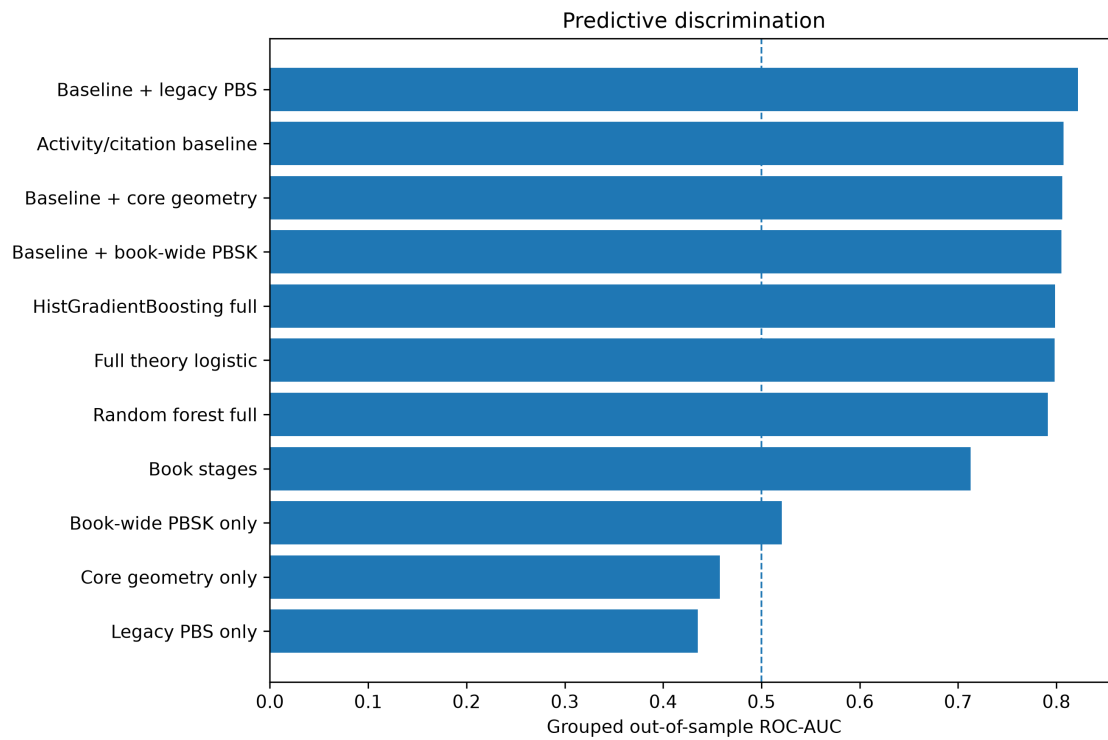


**Figure S11:** Historical cross-source comparison of T-1 book-wide PBSK scores for overlapping OpenAlex and PubMed regions. The dashed diagonal indicates exact cross-source agreement.





**Figure S12:** Retrieval coverage by source and temporal window, showing mean records per region for OpenAlex and PubMed in pre- and post-event retrievals.



**Figure S13:** Independent power-validation grouped out-of-sample ROC-AUC across all candidate models. The dashed vertical line marks chance discrimination at ROC-AUC = 0.5.

## Software-generated tables

Tables S1 to S34 provide a complete preview-level audit trail for all 34 software-generated result tables. Full machine-readable versions are retained in `supplementary_data/results_tables/`.

**Table S1:** Historical landmark-cohort ablation analysis, showing grouped predictive performance after removal of each book-derived component.

| removed component         | n  | n features | roc auc |
|---------------------------|----|------------|---------|
| none                      | 92 | 15         | 0.4997  |
| stage_possibility         | 92 | 14         | 0.5019  |
| stage_question            | 92 | 14         | 0.4964  |
| stage_pattern             | 92 | 14         | 0.4986  |
| stage_imagination         | 92 | 14         | 0.4975  |
| stage_insight             | 92 | 14         | 0.4986  |
| stage_discovery_readiness | 92 | 14         | 0.5085  |
| bookwide_pbsk             | 92 | 14         | 0.4986  |
| core_geometry_score       | 92 | 14         | 0.5151  |

*Note.* Preview shows 9 of 9 rows and 4 of 7 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_ablation_openalex.csv`.

**Table S2:** Complete 40-chapter operationalization linking the conceptual source to PBSK literature-level themes, proxies, and software measures.

| chapter | part                | title                          | book theme                |
|---------|---------------------|--------------------------------|---------------------------|
| 1       | The Invisible World | The Day Before the Idea        | Pre-existence of novelty  |
| 2       | The Invisible World | The Territory of Possibility   | Possibility space         |
| 3       | The Invisible World | The Things That Do Not Yet ... | Unrealized futures        |
| 4       | The Invisible World | The Invisible Door             | Anomalies/openings        |
| 5       | The Invisible World | The Shape of the Unseen        | Hidden structure          |
| 6       | The Invisible World | The First Impossible Thing     | Boundary challenge        |
| 7       | The Invisible World | The Library of Futures         | Future possibility space  |
| 8       | The Invisible World | Why Reality Is Incomplete      | Knowledge incompleteness  |
| 9       | The Question        | The Difference Between Seei... | Attention to weak signals |
| 10      | The Question        | The Question Nobody Asked      | Unasked questions         |

*Note.* Preview shows 10 of 40 rows and 4 of 7 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_book_operationalization.csv`.

**Table S3:** Exploratory generalization of predictive performance across the software-defined creative-universe realms represented in the landmark cohort.

| <b>model</b>               | <b>book realm</b> | <b>n</b> | <b>roc auc</b> |
|----------------------------|-------------------|----------|----------------|
| Activity/citation baseline | Life              | 47       | 0.5458         |
| Activity/citation baseline | Machines          | 34       | 0.428          |
| Activity/citation baseline | Nature            | 11       | 1              |
| Baseline + book-wide PBSK  | Life              | 47       | 0.5646         |
| Baseline + book-wide PBSK  | Machines          | 34       | 0.3788         |
| Baseline + book-wide PBSK  | Nature            | 11       | 1              |
| Baseline + core geometry   | Life              | 47       | 0.4917         |
| Baseline + core geometry   | Machines          | 34       | 0.3333         |
| Baseline + core geometry   | Nature            | 11       | 1              |
| Baseline + legacy PBS      | Life              | 47       | 0.4521         |

*Note.* Preview shows 10 of 33 rows and 4 of 6 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_creative_universe_generalization.csv`.

**Table S4:** Cross-source agreement between OpenAlex and PubMed for selected landmark-cohort PBSK features.

| <b>metric</b>        | <b>n</b> | <b>pearson r</b> | <b>spearman rho</b> |
|----------------------|----------|------------------|---------------------|
| bookwide_pbsk        | 10       | 0.1773           | 0.05455             |
| core_geometry_score  | 10       | 0.4946           | 0.6121              |
| skg                  | 10       | -0.4605          | -0.2242             |
| possibility_pressure | 10       | -0.3113          | -0.4667             |
| question_pressure    | 10       | -0.08048         | 0.06667             |
| collision_index      | 10       | 0.1724           | 0.0303              |
| insight_coherence    | 10       | -0.2096          | -0.05628            |
| sequence_order_score | 10       | -0.5818          | -0.6636             |

*Note.* Preview shows 8 of 8 rows and 4 of 5 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_cross_source_agreement.csv`.

**Table S5:** Quality-control status for every software-generated figure in the frozen PBSK v4.0.0 run.

| figure                         | status | message |
|--------------------------------|--------|---------|
| figure_01_benchmark_timeline   | OK     |         |
| figure_02_bookwide_pbsk_tra... | OK     |         |
| figure_03_book_stage_profil... | OK     |         |
| figure_04_bookwide_pbsk_dis... | OK     |         |
| figure_05_sequence_architec... | OK     |         |
| figure_06_predictive_perfor... | OK     |         |
| figure_07_temporal_prediction  | OK     |         |
| figure_08_bookwide_ablation    | OK     |         |
| figure_09_transformation       | OK     |         |
| figure_10_creative_universe... | OK     |         |

*Note.* Preview shows 10 of 14 rows and 3 of 3 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_figure_generation_qc.csv`.

**Table S6:** Complete OpenAlex landmark-cohort hypothesis-test output across confirmatory and exploratory families and temporal offsets.

| cohort               | offset | metric               | family       |
|----------------------|--------|----------------------|--------------|
| landmark_exploratory | 5      | skg                  | confirmatory |
| landmark_exploratory | 5      | collision_index      | confirmatory |
| landmark_exploratory | 5      | sequence_order_score | confirmatory |
| landmark_exploratory | 5      | bookwide_pbsk        | confirmatory |
| landmark_exploratory | 5      | core_geometry_score  | confirmatory |
| landmark_exploratory | 5      | pbs                  | exploratory  |
| landmark_exploratory | 5      | possibility_pressure | exploratory  |
| landmark_exploratory | 5      | question_pressure    | exploratory  |
| landmark_exploratory | 5      | anomaly_pressure     | exploratory  |
| landmark_exploratory | 5      | doubt_tension        | exploratory  |

*Note.* Preview shows 10 of 69 rows and 4 of 12 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_hypothesis_tests_openalex.csv`.

**Table S7:** Incremental predictive-value contrasts for the historical landmark cohort relative to the activity/citation baseline.

| baseline model             | augmented model           | n  | n groups |
|----------------------------|---------------------------|----|----------|
| Activity/citation baseline | Baseline + legacy PBS     | 92 | 12       |
| Activity/citation baseline | Baseline + book-wide PBSK | 92 | 12       |
| Activity/citation baseline | Baseline + core geometry  | 92 | 12       |
| Activity/citation baseline | Full theory logistic      | 92 | 12       |

*Note.* Preview shows 4 of 4 rows and 4 of 8 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_incremental_value_openalex.csv`.



**Table S8:** Historical landmark benchmark cases, search queries, domains, landmark years, and software metadata.

| case id     | label                          | query                          | domain            |
|-------------|--------------------------------|--------------------------------|-------------------|
| crispr      | CRISPR-Cas genome editing      | CRISPR Cas9 genome editing     | Life Sciences     |
| ips         | Induced pluripotent stem cells | induced pluripotent stem ce... | Life Sciences     |
| mrna        | mRNA vaccine technology        | messenger RNA vaccine mRNA     | Life Sciences     |
|             |                                | ...                            |                   |
| alphafold2  | AlphaFold2 protein structur... | protein structure predictio... | Life Sciences     |
| singlecell  | Single-cell RNA sequencing     | single cell RNA sequencing ... | Life Sciences     |
| rnai        | RNA interference               | RNA interference double str... | Life Sciences     |
| graphene    | Graphene isolation             | graphene two dimensional ca... | Physical Sciences |
| transformer | Transformer architecture       | attention transformer neura... | Computer Science  |
| resnet      | Deep residual networks         | deep residual learning resi... | Computer Science  |
| diffusion   | Diffusion generative models    | diffusion probabilistic gen... | Computer Science  |

*Note.* Preview shows 10 of 12 rows and 4 of 8 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_landmark_benchmark.csv`.

**Table S9:** Matched control regions used for the historical landmark benchmark cohort.

| control id    | match case id | label                          | query                          |
|---------------|---------------|--------------------------------|--------------------------------|
| crispr_c1     | crispr        | Zinc-finger nuclease engine... | zinc finger nuclease genome... |
| crispr_c2     | crispr        | Meganuclease genome enginee... | meganuclease genome enginee... |
| ips_c1        | ips           | Adult stem-cell transdiffer... | adult stem cell transdiffer... |
| ips_c2        | ips           | Somatic cell nuclear transfer  | somatic cell nuclear transf... |
| mrna_c1       | mrna          | DNA vaccine platforms          | DNA vaccine platform immuni... |
| mrna_c2       | mrna          | Viral-vector vaccine platforms | viral vector vaccine platform  |
| alphafold2_c1 | alphafold2    | Protein threading methods      | protein threading structure... |
| alphafold2_c2 | alphafold2    | Homology modelling             | protein homology modeling s... |
| singlecell_c1 | singlecell    | Bulk RNA sequencing            | bulk RNA sequencing transcr... |
| singlecell_c2 | singlecell    | Microarray transcriptomics     | microarray transcriptome ge... |

*Note.* Preview shows 10 of 24 rows and 4 of 6 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_matched_controls.csv`.

**Table S10:** Grouped out-of-fold predictions generated for the historical OpenAlex landmark cohort.

| region id  | match case id | offset | target |
|------------|---------------|--------|--------|
| crispr     | crispr        | 5      | 1      |
| crispr     | crispr        | 3      | 1      |
| crispr     | crispr        | 1      | 1      |
| ips        | ips           | 5      | 1      |
| ips        | ips           | 3      | 1      |
| ips        | ips           | 1      | 1      |
| mrna       | mrna          | 5      | 1      |
| mrna       | mrna          | 3      | 1      |
| mrna       | mrna          | 1      | 1      |
| alphafold2 | alphafold2    | 1      | 1      |

*Note.* Preview shows 10 of 1012 rows and 4 of 7 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_oof_predictions_openalex.csv`.

**Table S11:** OpenAlex book-wide feature matrix for the historical landmark and matched-control regions.

| source   | region id  | offset | cutoff year |
|----------|------------|--------|-------------|
| OpenAlex | crispr     | 5      | 2007        |
| OpenAlex | crispr     | 3      | 2009        |
| OpenAlex | crispr     | 1      | 2011        |
| OpenAlex | ips        | 5      | 2001        |
| OpenAlex | ips        | 3      | 2003        |
| OpenAlex | ips        | 1      | 2005        |
| OpenAlex | mrna       | 5      | 2015        |
| OpenAlex | mrna       | 3      | 2017        |
| OpenAlex | mrna       | 1      | 2019        |
| OpenAlex | alphafold2 | 5      | 2016        |

*Note.* Preview shows 10 of 108 rows and 4 of 50 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_openalex_bookwide_features.csv`.

**Table S12:** Ablation analysis for the independent power-validation cohort.

| removed component         | n    | n features | roc auc |
|---------------------------|------|------------|---------|
| none                      | 1080 | 15         | 0.8027  |
| stage_possibility         | 1080 | 14         | 0.79    |
| stage_question            | 1080 | 14         | 0.8075  |
| stage_pattern             | 1080 | 14         | 0.8046  |
| stage_imagination         | 1080 | 14         | 0.8024  |
| stage_insight             | 1080 | 14         | 0.804   |
| stage_discovery_readiness | 1080 | 14         | 0.8037  |
| bookwide_pbsk             | 1080 | 14         | 0.8063  |
| core_geometry_score       | 1080 | 14         | 0.8039  |

*Note.* Preview shows 9 of 9 rows and 4 of 7 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_power_validation_ablation.csv`.

**Table S13:** Matched control topics for the independent power-validation cohort, including pre-cutoff matching metadata.

| topic id                                                              | topic name                     | domain            | field                          |
|-----------------------------------------------------------------------|--------------------------------|-------------------|--------------------------------|
| <a href="https://openalex.org/T14341">https://openalex.org/T14341</a> | Education, Technology, and ... | Physical Sciences | Environmental Science          |
| <a href="https://openalex.org/T12643">https://openalex.org/T12643</a> | Sustainability and Ecologic... | Physical Sciences | Environmental Science          |
| <a href="https://openalex.org/T14054">https://openalex.org/T14054</a> | Humic Substances and Bio-Or... | Life Sciences     | Agricultural and Biological... |
| <a href="https://openalex.org/T13218">https://openalex.org/T13218</a> | Sesame and Sesamin Research    | Life Sciences     | Agricultural and Biological... |
| <a href="https://openalex.org/T14027">https://openalex.org/T14027</a> | 21st Century Education and ... | Social Sciences   | Social Sciences                |
| <a href="https://openalex.org/T13982">https://openalex.org/T13982</a> | Social Movements and Cultur... | Social Sciences   | Social Sciences                |
| <a href="https://openalex.org/T14407">https://openalex.org/T14407</a> | Athletic Training and Educa... | Health Sciences   | Health Professions             |
| <a href="https://openalex.org/T14447">https://openalex.org/T14447</a> | Trade Secret Protection Met... | Health Sciences   | Health Professions             |
| <a href="https://openalex.org/T13975">https://openalex.org/T13975</a> | Energy, Economy, and Techno... | Physical Sciences | Energy                         |
| <a href="https://openalex.org/T14374">https://openalex.org/T14374</a> | Statistical Methods in Epid... | Physical Sciences | Mathematics                    |

*Note.* Preview shows 10 of 240 rows and 4 of 17 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_power_validation_controls.csv`.

**Table S14:** Cross-source agreement between OpenAlex and PubMed features in the power-validation overlap.

| <b>metric</b>        | <b>n</b> | <b>pearson r</b> | <b>spearman rho</b> |
|----------------------|----------|------------------|---------------------|
| bookwide_pbsk        | 79       | 0.09993          | -0.04734            |
| core_geometry_score  | 79       | -0.02655         | -0.04575            |
| skg                  | 79       | 0.04281          | -0.02018            |
| possibility_pressure | 79       | 0.07772          | 0.1397              |
| question_pressure    | 79       | 0.1109           | 0.3472              |
| collision_index      | 79       | -0.009221        | -0.02438            |
| insight_coherence    | 79       | 0.04573          | 0.004366            |
| sequence_order_score | 79       | 0.168            | 0.1482              |

*Note.* Preview shows 8 of 8 rows and 4 of 5 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_power_validation_cross_source_agreement.csv`.

**Table S15:** OpenAlex feature matrix for the independent power-validation cohort across temporal offsets.

| source   | region id  | offset | cutoff year |
|----------|------------|--------|-------------|
| OpenAlex | PV_0000_P  | 5      | 2014        |
| OpenAlex | PV_0000_P  | 3      | 2016        |
| OpenAlex | PV_0000_P  | 1      | 2018        |
| OpenAlex | PV_0000_C1 | 5      | 2014        |
| OpenAlex | PV_0000_C1 | 3      | 2016        |
| OpenAlex | PV_0000_C1 | 1      | 2018        |
| OpenAlex | PV_0000_C2 | 5      | 2014        |
| OpenAlex | PV_0000_C2 | 3      | 2016        |
| OpenAlex | PV_0000_C2 | 1      | 2018        |
| OpenAlex | PV_0001_P  | 5      | 2014        |

*Note.* Preview shows 10 of 1080 rows and 4 of 50 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_power_validation_features_openalex.csv`.



**Table S16:** PubMed feature matrix retrieved for power-validation regions with available biomedical coverage.

| source | region id  | offset | cutoff year |
|--------|------------|--------|-------------|
| PubMed | PV_0001_P  | 5      | 2014        |
| PubMed | PV_0001_P  | 3      | 2016        |
| PubMed | PV_0001_P  | 1      | 2018        |
| PubMed | PV_0001_C1 | 5      | 2014        |
| PubMed | PV_0001_C1 | 3      | 2016        |
| PubMed | PV_0001_C1 | 1      | 2018        |
| PubMed | PV_0001_C2 | 5      | 2014        |
| PubMed | PV_0001_C2 | 3      | 2016        |
| PubMed | PV_0001_C2 | 1      | 2018        |
| PubMed | PV_0003_C1 | 5      | 2014        |

*Note.* Preview shows 10 of 354 rows and 4 of 50 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_power_validation_features_pubmed.csv`.

**Table S17:** Complete hypothesis-test output for the independent power-validation cohort, including confirmatory and exploratory families.

| cohort                        | offset | metric               | family       |
|-------------------------------|--------|----------------------|--------------|
| power_validation_confirmatory | 5      | skg                  | confirmatory |
| power_validation_confirmatory | 5      | collision_index      | confirmatory |
| power_validation_confirmatory | 5      | sequence_order_score | confirmatory |
| power_validation_confirmatory | 5      | bookwide_pbsk        | confirmatory |
| power_validation_confirmatory | 5      | core_geometry_score  | confirmatory |
| power_validation_confirmatory | 5      | pbs                  | exploratory  |
| power_validation_confirmatory | 5      | possibility_pressure | exploratory  |
| power_validation_confirmatory | 5      | question_pressure    | exploratory  |
| power_validation_confirmatory | 5      | anomaly_pressure     | exploratory  |
| power_validation_confirmatory | 5      | doubt_tension        | exploratory  |

*Note.* Preview shows 10 of 69 rows and 4 of 12 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_power_validation_hypotheses.csv`.

**Table S18:** Incremental predictive-value contrasts in the independent power-validation cohort relative to the activity/citation baseline.

| baseline model             | augmented model           | n    | n groups |
|----------------------------|---------------------------|------|----------|
| Activity/citation baseline | Baseline + legacy PBS     | 1080 | 120      |
| Activity/citation baseline | Baseline + book-wide PBSK | 1080 | 120      |
| Activity/citation baseline | Baseline + core geometry  | 1080 | 120      |
| Activity/citation baseline | Full theory logistic      | 1080 | 120      |

*Note.* Preview shows 4 of 4 rows and 4 of 8 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_power_validation_incremental_value.csv`.

**Table S19:** Grouped out-of-fold predictions for all candidate models in the independent power-validation cohort.

| region id  | match case id | offset | target |
|------------|---------------|--------|--------|
| PV_0000_P  | PV_0000       | 5      | 1      |
| PV_0000_P  | PV_0000       | 3      | 1      |
| PV_0000_P  | PV_0000       | 1      | 1      |
| PV_0000_C1 | PV_0000       | 5      | 0      |
| PV_0000_C1 | PV_0000       | 3      | 0      |
| PV_0000_C1 | PV_0000       | 1      | 0      |
| PV_0000_C2 | PV_0000       | 5      | 0      |
| PV_0000_C2 | PV_0000       | 3      | 0      |
| PV_0000_C2 | PV_0000       | 1      | 0      |
| PV_0001_P  | PV_0001       | 5      | 1      |

*Note.* Preview shows 10 of 11880 rows and 4 of 7 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_power_validation_oof_predictions.csv`.

**Table S20:** Future-emergent OpenAlex topics selected as positive cases in the independent power-validation cohort.

| topic id                                                              | topic name                     | domain            | field                          |
|-----------------------------------------------------------------------|--------------------------------|-------------------|--------------------------------|
| <a href="https://openalex.org/T11926">https://openalex.org/T11926</a> | Subterranean biodiversity a... | Physical Sciences | Earth and Planetary Sciences   |
| <a href="https://openalex.org/T12710">https://openalex.org/T12710</a> | Engineering and Agricultura... | Life Sciences     | Agricultural and Biological... |
| <a href="https://openalex.org/T14443">https://openalex.org/T14443</a> | COVID-19, Geopolitics, Tech... | Social Sciences   | Social Sciences                |
| <a href="https://openalex.org/T13945">https://openalex.org/T13945</a> | Legal, Health, Environmenta... | Health Sciences   | Medicine                       |
| <a href="https://openalex.org/T13647">https://openalex.org/T13647</a> | AI and Big Data Applications   | Physical Sciences | Computer Science               |
| <a href="https://openalex.org/T14182">https://openalex.org/T14182</a> | Technology-Enhanced Educati... | Social Sciences   | Social Sciences                |
| <a href="https://openalex.org/T11396">https://openalex.org/T11396</a> | Artificial Intelligence in ... | Health Sciences   | Health Professions             |
| <a href="https://openalex.org/T11307">https://openalex.org/T11307</a> | Domain Adaptation and Few-S... | Physical Sciences | Computer Science               |
| <a href="https://openalex.org/T11101">https://openalex.org/T11101</a> | Online Learning Methods and... | Social Sciences   | Social Sciences                |
| <a href="https://openalex.org/T12702">https://openalex.org/T12702</a> | Brain Tumor Detection and C... | Life Sciences     | Neuroscience                   |

*Note.* Preview shows 10 of 120 rows and 4 of 16 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_power_validation_positive_topics.csv`.

**Table S21:** Grouped out-of-sample predictive performance and bootstrap uncertainty for the independent power-validation cohort.

| <b>model</b>               | <b>n</b> | <b>n groups</b> | <b>n features</b> |
|----------------------------|----------|-----------------|-------------------|
| Activity/citation baseline | 1080     | 120             | 7                 |
| Legacy PBS only            | 1080     | 120             | 1                 |
| Book-wide PBSK only        | 1080     | 120             | 1                 |
| Core geometry only         | 1080     | 120             | 1                 |
| Book stages                | 1080     | 120             | 6                 |
| Baseline + legacy PBS      | 1080     | 120             | 14                |
| Baseline + book-wide PBSK  | 1080     | 120             | 8                 |
| Baseline + core geometry   | 1080     | 120             | 8                 |
| Full theory logistic       | 1080     | 120             | 36                |
| Random forest full         | 1080     | 120             | 36                |

*Note.* Preview shows 10 of 11 rows and 4 of 14 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_power_validation_predictive_performance.csv`.

**Table S22:** PubMed confirmatory replication output for the independent power-validation cohort. The frozen run produced no valid rows for this analysis.

---

The frozen run produced an empty machine-readable table for `table_power_validation_pubmed_replication.csv`. The zero-row CSV is retained under `supplementary_data/results_tables/` for provenance and to make the absence of a valid replicated test family explicit.

---

**Table S23:** Temporal predictive performance in the independent power-validation cohort at T-5, T-3, and T-1.

| offset | model                      | n   | n groups |
|--------|----------------------------|-----|----------|
| 5      | Activity/citation baseline | 360 | 120      |
| 5      | Book-wide PBSK only        | 360 | 120      |
| 5      | Core geometry only         | 360 | 120      |
| 5      | Baseline + core geometry   | 360 | 120      |
| 5      | Full theory logistic       | 360 | 120      |
| 3      | Activity/citation baseline | 360 | 120      |
| 3      | Book-wide PBSK only        | 360 | 120      |
| 3      | Core geometry only         | 360 | 120      |
| 3      | Baseline + core geometry   | 360 | 120      |
| 3      | Full theory logistic       | 360 | 120      |

*Note.* Preview shows 10 of 15 rows and 4 of 15 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_power_validation_temporal_performance.csv`.



**Table S24:** Grouped out-of-sample predictive performance for the historical OpenAlex landmark cohort.

| model                      | n  | n groups | n features |
|----------------------------|----|----------|------------|
| Activity/citation baseline | 92 | 12       | 7          |
| Legacy PBS only            | 92 | 12       | 1          |
| Book-wide PBSK only        | 92 | 12       | 1          |
| Core geometry only         | 92 | 12       | 1          |
| Book stages                | 92 | 12       | 6          |
| Baseline + legacy PBS      | 92 | 12       | 14         |
| Baseline + book-wide PBSK  | 92 | 12       | 8          |
| Baseline + core geometry   | 92 | 12       | 8          |
| Full theory logistic       | 92 | 12       | 36         |
| Random forest full         | 92 | 12       | 36         |

*Note.* Preview shows 10 of 11 rows and 4 of 14 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_predictive_performance_openalex.csv`.

**Table S25:** PubMed book-wide feature matrix for historical landmark regions with available biomedical coverage.

| source | region id  | offset | cutoff year |
|--------|------------|--------|-------------|
| PubMed | ips        | 5      | 2001        |
| PubMed | ips        | 3      | 2003        |
| PubMed | ips        | 1      | 2005        |
| PubMed | mrna       | 5      | 2015        |
| PubMed | mrna       | 3      | 2017        |
| PubMed | mrna       | 1      | 2019        |
| PubMed | alphafold2 | 5      | 2016        |
| PubMed | alphafold2 | 3      | 2018        |
| PubMed | alphafold2 | 1      | 2020        |
| PubMed | singlecell | 5      | 2004        |

*Note.* Preview shows 10 of 48 rows and 4 of 50 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_pubmed_bookwide_features.csv`.

**Table S26:** PubMed landmark-cohort replication hypothesis output. The frozen run produced no valid rows for this analysis.

---

The frozen run produced an empty machine-readable table for `table_pubmed_replication_hypotheses.csv`. The zero-row CSV is retained under `supplementary_data/results_tables/` for provenance and to make the absence of a valid replicated test family explicit.

---

**Table S27:** PubMed post-landmark transformation replication results for available biomedical cases.

| <b>metric</b>       | <b>n pairs</b> | <b>positive mean</b> | <b>control mean</b> |
|---------------------|----------------|----------------------|---------------------|
| post_growth         | 6              | 0.8578               | 0.7425              |
| topic_diffusion     | 6              | 0.5868               | 0.4704              |
| normalization_index | 6              | 0.7508               | 0.5793              |
| post_adoption       | 6              | 0                    | 0                   |
| transformation      | 6              | 0.5489               | 0.448               |

*Note.* Preview shows 5 of 5 rows and 4 of 9 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_pubmed_replication_transformation.csv`.

**Table S28:** Combined region-level feature matrix across OpenAlex and PubMed sources and study cohorts.

| source   | region id  | offset | cutoff year |
|----------|------------|--------|-------------|
| OpenAlex | crispr     | 5      | 2007        |
| OpenAlex | crispr     | 3      | 2009        |
| OpenAlex | crispr     | 1      | 2011        |
| OpenAlex | ips        | 5      | 2001        |
| OpenAlex | ips        | 3      | 2003        |
| OpenAlex | ips        | 1      | 2005        |
| OpenAlex | mrna       | 5      | 2015        |
| OpenAlex | mrna       | 3      | 2017        |
| OpenAlex | mrna       | 1      | 2019        |
| OpenAlex | alphafold2 | 5      | 2016        |

*Note.* Preview shows 10 of 1590 rows and 4 of 50 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_region_features_all_sources.csv`.

**Table S29:** Retrieval quality-control record by source, region, temporal window, and query status.

| source   | cohort   | region id  | match case id |
|----------|----------|------------|---------------|
| OpenAlex | landmark | crispr     | crispr        |
| OpenAlex | landmark | crispr     | crispr        |
| OpenAlex | landmark | ips        | ips           |
| OpenAlex | landmark | ips        | ips           |
| OpenAlex | landmark | mrna       | mrna          |
| OpenAlex | landmark | mrna       | mrna          |
| OpenAlex | landmark | alphafold2 | alphafold2    |
| OpenAlex | landmark | alphafold2 | alphafold2    |
| OpenAlex | landmark | singlecell | singlecell    |
| OpenAlex | landmark | singlecell | singlecell    |

*Note.* Preview shows 10 of 1068 rows and 4 of 12 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_retrieval_qc.csv`.

**Table S30:** Historical landmark-cohort sequence-order architecture across T-5, T-3, and T-1.

| offset | class        | n  | mean sequence order score |
|--------|--------------|----|---------------------------|
| 5      | breakthrough | 8  | 0.4458                    |
| 5      | control      | 19 | 0.4246                    |
| 3      | breakthrough | 10 | 0.3833                    |
| 3      | control      | 22 | 0.4076                    |
| 1      | breakthrough | 11 | 0.3879                    |
| 1      | control      | 22 | 0.4242                    |

*Note.* Preview shows 6 of 6 rows and 4 of 5 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_sequence_architecture.csv`.

**Table S31:** Temporal leakage audit for all region-level feature rows used or retained by the frozen run.

| source   | region id  | offset | cutoff year |
|----------|------------|--------|-------------|
| OpenAlex | crispr     | 5      | 2007        |
| OpenAlex | crispr     | 3      | 2009        |
| OpenAlex | crispr     | 1      | 2011        |
| OpenAlex | ips        | 5      | 2001        |
| OpenAlex | ips        | 3      | 2003        |
| OpenAlex | ips        | 1      | 2005        |
| OpenAlex | mrna       | 5      | 2015        |
| OpenAlex | mrna       | 3      | 2017        |
| OpenAlex | mrna       | 1      | 2019        |
| OpenAlex | alphafold2 | 5      | 2016        |

*Note.* Preview shows 10 of 1590 rows and 4 of 6 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_temporal_leakage_audit.csv`.



**Table S32:** Temporal predictive performance for the historical OpenAlex landmark cohort.

| offset | model                      | n  | n groups |
|--------|----------------------------|----|----------|
| 5      | Activity/citation baseline | 27 | 12       |
| 5      | Book-wide PBSK only        | 27 | 12       |
| 5      | Core geometry only         | 27 | 12       |
| 5      | Baseline + core geometry   | 27 | 12       |
| 5      | Full theory logistic       | 27 | 12       |
| 3      | Activity/citation baseline | 32 | 12       |
| 3      | Book-wide PBSK only        | 32 | 12       |
| 3      | Core geometry only         | 32 | 12       |
| 3      | Baseline + core geometry   | 32 | 12       |
| 3      | Full theory logistic       | 32 | 12       |

*Note.* Preview shows 10 of 15 rows and 4 of 15 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_temporal_predictive_performance.csv`.

**Table S33:** Region-level post-landmark transformation outcomes, including growth, diffusion, normalization, adoption, and composite transformation measures.

| source   | cohort   | region id   | match case id |
|----------|----------|-------------|---------------|
| OpenAlex | landmark | crispr      | crispr        |
| OpenAlex | landmark | ips         | ips           |
| OpenAlex | landmark | mrna        | mrna          |
| OpenAlex | landmark | alphafold2  | alphafold2    |
| OpenAlex | landmark | singlecell  | singlecell    |
| OpenAlex | landmark | rnai        | rnai          |
| OpenAlex | landmark | graphene    | graphene      |
| OpenAlex | landmark | transformer | transformer   |
| OpenAlex | landmark | resnet      | resnet        |
| OpenAlex | landmark | diffusion   | diffusion     |

*Note.* Preview shows 10 of 414 rows and 4 of 15 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_transformation_outcomes.csv`.

**Table S34:** Paired OpenAlex post-landmark transformation tests for the historical landmark cohort.

| <b>metric</b>       | <b>n pairs</b> | <b>positive mean</b> | <b>control mean</b> |
|---------------------|----------------|----------------------|---------------------|
| post_growth         | 12             | 0.6401               | 0.5762              |
| topic_diffusion     | 12             | 0.4567               | 0.4954              |
| normalization_index | 12             | 0.6143               | 0.5559              |
| post_adoption       | 12             | 0.5599               | 0.564               |
| transformation      | 12             | 0.5678               | 0.5479              |

*Note.* Preview shows 5 of 5 rows and 4 of 9 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_transformation_tests_openalex.csv`.

## **Operationalization and software provenance**

The computational formulas and configuration underlying these tables are described in the main manuscript; the book-derived conceptual arc is cited as the motivating source (Akhtar, 2026). The exact frozen software files used for the analysis are included under `supplementary_data/software_snapshot/`, including the feature-engineering, modeling, statistics, power-validation, and configuration files. This snapshot is included to ensure that the numerical tables can be traced to executable definitions rather than prose alone.

## **References cited in the supplement**

Akhtar, M. A. K. (2026). *The impossible idea: How new realities enter the world* [Published under the pen name Shunya]. <https://a.co/d/03buY0qI>